

CarDiff — Research Spotlight

60-second speaking script • MIUA 2026

Target $\approx$ 60 s • $\approx$ 150 words • $\approx$ 150 wpm

Read one block per slide. **Bold** = light vocal stress. Advance the slide at each time marker.

Slide 1 • Title & Authors

0:00–0:06 • $\sim$ 14 w

Hi, I'm **Sanyam Jain** from **Aarhus University**. [pause] With my co-authors, I'm glad to present **CarDiff**.

Slide 2 • Motivation

0:06–0:22 • $\sim$ 44 w

Dental caries is the **most common disease in the world**, and catching it **early** on X-rays really matters. But AI for caries detection is held back by two problems: **too few annotated images**, and **severe class imbalance** — the early lesions we most want to catch are the **rarest**.

Slide 3 • Technical Innovation

0:22–0:42 • $\sim$ 50 w

Our solution is **CarDiff** — a segmentation-guided **latent diffusion** model that synthesizes realistic dental radiographs from lesion masks. [pause] It learns through **two paths**, paired and self-supervised, with **anatomy-aware conditioning**: a patch-graph network and global attention, injected via FiLM. CarDiff reaches the **best FID and LPIPS** of four baselines.

Slide 4 • Potential Impact

0:42–1:00 • $\sim$ 48 w

So why does it matter? [pause] Using CarDiff images to augment training **five-fold** improves caries segmentation across **seven architectures** — up to **six-and-a-half percent** higher Dice, with the biggest gains on the **rare early lesions**. It's a step toward **trustworthy, AI-assisted diagnosis** when data is scarce. [pause] Thank you.

Delivery tips

- Breathe and **pause** at each [pause] — it resets the clock and adds emphasis.
- Land the three numbers clearly: **best FID/LPIPS**, **5 $\times$** , **+6.5% Dice**.
- Open warm on Slide 1, lift energy on Slide 2 (the hook), slow slightly on “Thank you.”
- If you run long, shorten the opening to just “I'm Sanyam Jain — here's CarDiff.”